

5. The following table shows the solubility of potassium permanganate in water at temperatures between 30 °C and 60 °C.

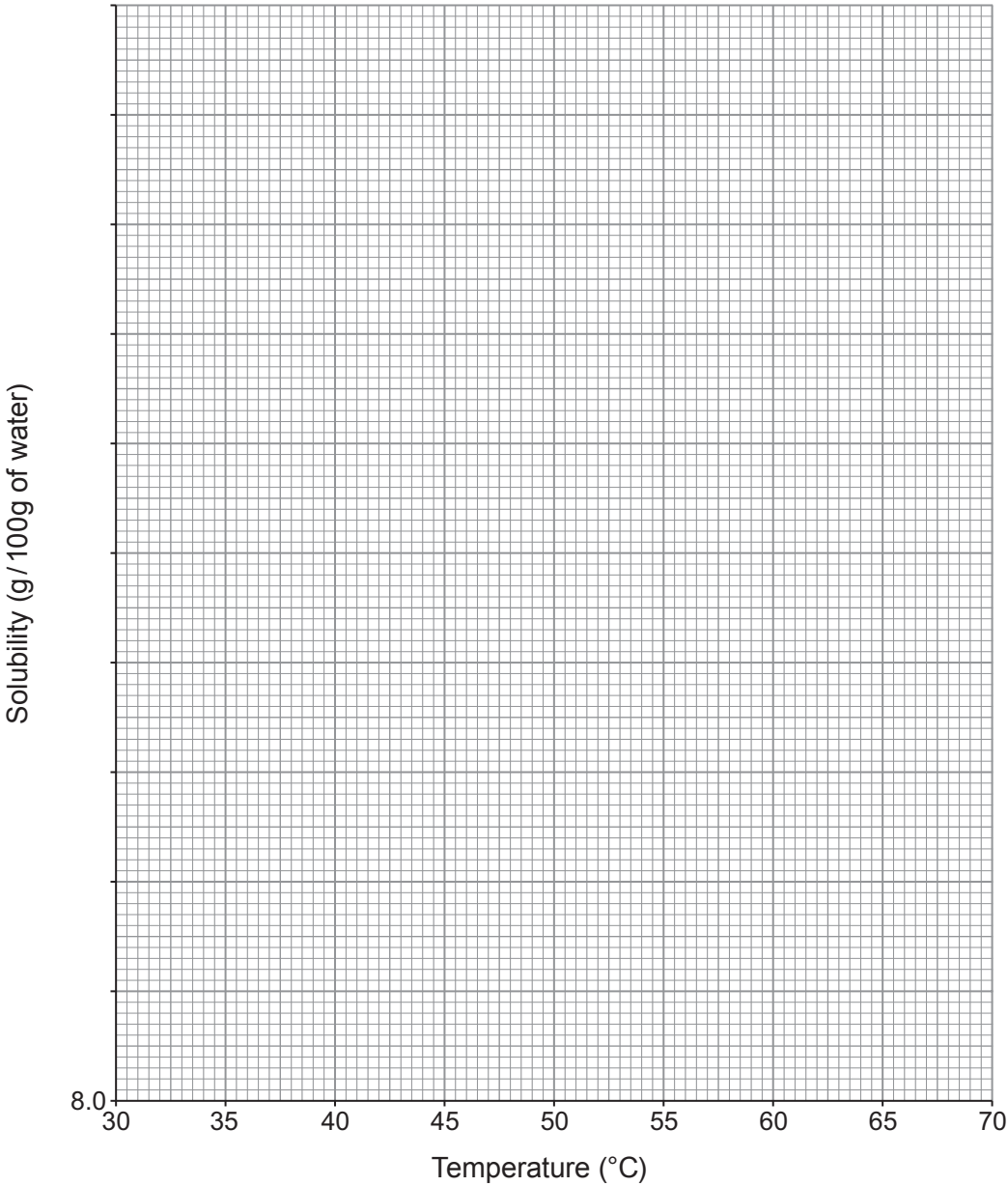
Temperature (°C)	Solubility (g / 100 g of water)
30	9.0
35	10.8
40	12.5
45	14.4
50	16.8
55	19.2
60	22.2



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(a) Plot these values on the grid below. Draw a suitable line.

[4]



(b) Use the graph to calculate the mass of crystals formed when a saturated solution in 500 g of water is cooled from 65 °C to 30 °C. [3]

Mass of crystals = g

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